Python: exercises on iteration

This notebook was generated in **pseudocode** mode.

Exercise 1: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):
    # Step 1: Set up a for loop to iterate from 1 to n (inclusive)
    # Step 2: For each number in the range, print it
```

Test your solution

```
import contextlib
import io
```

```python
captured_output = io.StringIO()
with contextlib.redirect_stdout(captured_output):
    print_numbers(5)

result = [int(line.strip()) for line in captured_output.getvalue().strip().split('\n'
assert result == [1, 2, 3, 4, 5]
print("✓ print_numbers tests passed")
```

```
✓ print_numbers tests passed
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 2: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers (1 + 2 + 3 + ... + N) using a while loop.

Your solution

```python
def sum_natural_numbers(n):
    # Step 1: Initialize sum variable to 0 and counter to 1
    # Step 2: Set up while loop that continues while counter <= n
    # Step 3: Add current counter value to sum
    # Step 4: Increment counter by 1
    # Step 5: Return the final sum
```

Test your solution

```python
# Test case 2: Test with another number
result = sum_natural_numbers(3)
expected = 1 + 2 + 3  # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```python
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}")  # 15
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 3: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():
    # Step 1: Set up a for loop with start=2, stop=21, step=2
    # Step 2: For each even number in the range, print it
```

Test your solution

```
import contextlib
import io

captured_output = io.StringIO()
with contextlib.redirect_stdout(captured_output):
    print_even_numbers()

result = [int(line.strip()) for line in captured_output.getvalue().strip().split('\n'
assert result == [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
print("✓ print_even_numbers tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 4: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):
    # Step 1: Check if n is even using modulo operator
    # Step 2: If n is even, set up while loop that continues while n > 0
    # Step 3: Print current n value
    # Step 4: Decrement n by 1
    # Step 5: If n is odd, do nothing
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)
countdown_if_even(6)  # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 5: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():
    # Step 1: Set up a for loop to iterate from 1 to 10
    # Step 2: For each multiplier, calculate 3 * multiplier
    # Step 3: Print the multiplication equation and result
```

Test your solution

```
import contextlib
import io

captured_output = io.StringIO()
with contextlib.redirect_stdout(captured_output):
    multiplication_table_3()

result = [line.strip() for line in captured_output.getvalue().strip().split('\n') if
expected = [f"3 x {i} = {3 * i}" for i in range(1, 11)]
assert result == expected
print("✓ multiplication_table_3 tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 6: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):
    # Step 1: Initialize sum to 0, current even number to 2, and count to 0
    # Step 2: Set up while loop that continues while count < n
    # Step 3: Add current even number to sum
    # Step 4: Move to next even number (add 2)
    # Step 5: Increment count
    # Step 6: Return the final sum
```

Test your solution

```
# Test case 2: Test with another number
result = sum_first_n_even(5)
expected = 2 + 4 + 6 + 8 + 10  # 30
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 7: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```python
def print_divisible_by_5():
    # Step 1: Set up a for loop with start=5, stop=51, step=5
    # Step 2: For each number divisible by 5, print it
```

Test your solution

```python
import contextlib
import io

captured_output = io.StringIO()
with contextlib.redirect_stdout(captured_output):
    print_divisible_by_5()

result = [int(line.strip()) for line in captured_output.getvalue().strip().split('\n'
```

```
assert result == [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
print("✓ print_divisible_by_5 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 8: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():
    # Step 1: Initialize factorial to 1 and counter to 5
    # Step 2: Set up while loop that continues while counter > 0
    # Step 3: Multiply factorial by current counter value
    # Step 4: Decrement counter by 1
    # Step 5: Return the final factorial
```

Test your solution

```
# Test case 2: Verify result
result = factorial_5()
assert result == 120, f"Expected 120, got {result}"
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 9: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():
    # Step 1: Initialize sum to 0
```

```
    # Step 2: Set up for loop with start=1, stop=21, step=2 (odd numbers)
    # Step 3: Add each odd number to sum
    # Step 4: Return the final sum
```

Test your solution

```
# Test case 2: Verify result
result = sum_odd_numbers()
assert result == 100, f"Expected 100, got {result}"
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_odd_numbers()
print(f"Sum of odd numbers 1-20: {result}")  # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 10: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):
    # Step 1: Set up for loop with start=k, stop=n+1, step=k
    # Step 2: For each number divisible by k, print it
```

Test your solution

```
# Test case 2: Test with n=20, k=4
# Expected: prints 4, 8, 12, 16, 20
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution
print_divisible_by_k(20, 3)  # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 11: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```
def count_digits(num):
    # Step 1: Initialize a counter to 0
    # Step 2: While num > 0, do integer division by 10 and increment counter
    # Step 3: Return the counter
```

Test your solution

```
assert count_digits(123456) == 6
assert count_digits(7) == 1
assert count_digits(10) == 2
print("✓ count_digits tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 12: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():
    # Step 1: Initialize an empty list
    # Step 2: For i from 1 to 5 inclusive, append i**3 to the list
    # Step 3: Return the list
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 13: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):
    # Step 1: Initialize product to 1
    # Step 2: For i from 1 to n inclusive, multiply product by i
    # Step 3: Return product
```

Test your solution

```
assert product_1_to_n(1) == 1
assert product_1_to_n(3) == 6
assert product_1_to_n(5) == 120
print("✓ product_1_to_n tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 14: Odd Numbers from 1 to 15

Exercise Description

Write a function that prints all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():
    # Step 1: Initialize i to 1
    # Step 2: While i <= 15, print i then increment by 2
```

Test your solution

```
import contextlib
import io
```

```python
# Capture the printed output
captured_output = io.StringIO()
with contextlib.redirect_stdout(captured_output):
    odd_1_to_15()

# Get the output and convert to list of integers for testing
output_lines = captured_output.getvalue().strip().split('\n')
result = [int(line.strip()) for line in output_lines if line.strip()]

assert result == [1, 3, 5, 7, 9, 11, 13, 15]
print("✓ odd_1_to_15 tests passed")
print(f"Generated odd numbers: {result}")
```

Debug your solution

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```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 15: Print Numbers from 5 to 1

Exercise Description

Write a function that prints numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():
    # Step 1: Use range from 5 down to 1 inclusive and print each number
```

Test your solution

```
import contextlib
import io

# Capture the printed output
captured_output = io.StringIO()
with contextlib.redirect_stdout(captured_output):
    countdown_5_to_1()

# Get the output and convert to list of integers for testing
output_lines = captured_output.getvalue().strip().split('\n')
result = [int(line.strip()) for line in output_lines if line.strip()]

assert result == [5, 4, 3, 2, 1]
print("✓ countdown_5_to_1 tests passed")
print(f"Generated countdown: {result}")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 16: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```python
def sum_first_7_odds():
    # Step 1: Initialize i=1, count=0, total=0
    # Step 2: While count < 7, add i to total, increment i by 2 and count by 1
    # Step 3: Return total
```

Test your solution

```python
assert sum_first_7_odds() == 49
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 17: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a concatenated string with the original string repeated as many times as its length. Use a loop and remember that `len(s)` gives you the length of string `s`.

Your solution

```python
def repeat_string_len_times(s):
    # Step 1: Initialize an empty result string
    # Step 2: For i in range(len(s)), concatenate s to result
    # Step 3: Return the concatenated string
```

Test your solution

```python
assert repeat_string_len_times("abc") == "abcabcabc"
assert repeat_string_len_times("") == ""
assert repeat_string_len_times("x") == "x"
print("✓ repeat_string_len_times tests passed")
```


Debug your solution